Question 1
Given $u(x,0) = x^2$, we want

$$u_t = u(x,0)_{xx} = 2$$

Integrating with respect to t, we get $u = 2t + C$. Since $u(x,0) = x^2$, we take $C = 0$. So we have

$$u(x,t) = x^2 + 2t$$

Question 2

(a) Given $u_t = Du_{xx} + f$. We have u_t meaning the rate of change in temperature of the rod over time, Du_{xx} means the heat flux of the rod, and f means the heat source.

(b) (i) Given $f(x) = 1$ with BCs $u(0,t) = 0$, $u(l,t) = 1$

$$u_t = Du_{xx} + 1$$

Since the steady state solution is time independent, $u_t = 0$

$$\begin{aligned} Du_{xx} &= -1 \\ u_{xx} &= -\frac{1}{D} \\ u_x &= -\frac{1}{D}x + f(t) \\ u &= -\frac{x^2}{2D} + ax + b \end{aligned}$$

Using BCs $u(0,t) = 0$, $u(l,t) = 1$

$$\begin{aligned} u(0) &= -\frac{0^2}{2D} + a \cdot 0 + b \Rightarrow b = 0 \\ u(l) &= -\frac{l^2}{2D} + al = 0 \\ \Rightarrow \frac{l^2}{2D} &= al \\ \frac{l}{2D} &= a \\ \Rightarrow u(x) &= -\frac{x^2}{2D} + \frac{lx}{2D} \end{aligned}$$

(ii) Given $f(x) = \begin{cases} 0 & 0 < x < l/2 \\ H & l/2 < x < l \end{cases}$ with BCs $u(0,t) = 0, u(l,t) = 0$

We have $Du_{xx} = 0$ for $0 < x < l/2$

$$\begin{aligned} Du_{xx} &= 0 \\ Du_x &= a \\ Du &= ax + b \\ u &= \frac{a}{D}x + \frac{b}{D} \end{aligned}$$

and $Du_{xx} = -H$ for $l/2 < x < l$

$$\begin{aligned} Du_{xx} &= -H \\ Du_x &= -Hx + c \\ Du &= -\frac{Hx^2}{2} + cx + e \\ u &= -\frac{Hx^2}{2D} + \frac{c}{D}x + \frac{e}{D} \end{aligned}$$

With our BCs, we have

$$\begin{aligned} u(0) = \frac{a}{D} \cdot 0 + \frac{b}{D} &= 0 \\ \Rightarrow \frac{b}{D} &= 0 \\ u(l) = -\frac{Hl^2}{2D} + \frac{c}{D}l + \frac{e}{D} &= 0 \\ \Rightarrow \frac{e}{D} &= \frac{Hl^2}{2D} - \frac{c}{D}l \end{aligned}$$

Considering the rod is continuous, we have two more constraints, where heat $u(x)$ and flux $-Du_x$ are continuous at $\frac{l}{2}$. With heat, we have

$$\begin{aligned} u(\frac{l}{2}-) &= u(\frac{l}{2}+) \\ \frac{al}{2D} &= -\frac{Hl^2}{8D} + \frac{cl}{2D} + \frac{Hl^2}{2D} - \frac{cl}{D} \\ \frac{al}{2D} &= \frac{3Hl^2}{8D} - \frac{cl}{2D} \\ \frac{a}{D} &= \frac{3Hl}{4D} - \frac{c}{D} \end{aligned}$$

With heat flux, we have

$$\begin{aligned} -Du_x(\frac{l}{2}-) &= -Du_x(\frac{l}{2}+) \\ u_x(\frac{l}{2}-) &= u_x(\frac{l}{2}+) \\ \frac{a}{D} &= -\frac{Hl}{2D} + \frac{c}{D} \end{aligned}$$

Combining the two, we get

$$\begin{aligned} \frac{3Hl}{4D} - \frac{c}{D} &= -\frac{Hl}{2D} + \frac{c}{D} \\ \frac{5Hl}{4D} &= \frac{2c}{D} \\ \frac{5Hl}{8D} &= \frac{c}{D} \\ \Rightarrow \frac{a}{D} &= -\frac{Hl}{2D} + \frac{5Hl}{8D} \\ &= \frac{Hl}{8D} \\ \Rightarrow \frac{e}{D} &= \frac{Hl^2}{2D} - \frac{5Hl^2}{8D} \\ &= -\frac{Hl^2}{8D} \end{aligned}$$

And so our solutions are

$$\begin{cases} u = \frac{Hl}{8D}x & 0 < x < \frac{l}{2} \\ u = -\frac{Hx^2}{2D} + \frac{5Hl}{8D}x - \frac{Hl^2}{8D} & \frac{l}{2} < x < l \end{cases}$$

Question 3

(a) $u_t = Du_{xx} - cu_x$

$$\begin{aligned} 0 &= Du_{xx} - cu_x \\ Du_{xx} &= cu_x \\ u_{xx} &= \frac{c}{D}u_x \end{aligned}$$

Let $v = u_x$

$$\begin{aligned}
v_x &= \frac{c}{D}v \\
v &= Ae^{\frac{c}{D}x} \\
u_x &= Ae^{\frac{c}{D}x} \\
u &= Ae^{\frac{c}{D}x} \cdot \frac{D}{c} + B \\
u &= Ae^{\frac{c}{D}x} + B
\end{aligned}$$

(b) $u_t = Du_{xx} - cu_x + ru$

$$0 = Du_{xx} - cu_x + ru$$

Question 4